Quasi-Newton Methods

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From the source, this section takes the order of ideas: the cost of Newton's method, the secant condition, and the BFGS and limited-memory BFGS updates. The source states the updates as solutions of nearest-matrix problems without proof; here the BFGS update is written out. The Davidon-Fletcher-Powell and rank-1 updates, the nearest-matrix derivation, the history, and the Julia code are not part of this section.

Definition 0.1 (Secant condition)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, and let $\underset{\sim}{x}, \underset{\sim}{x}^+ \in \mathbb{R}^p$. Write $\underset{\sim}{s} \stackrel{\text{def}}{=} \underset{\sim}{x}^+ - \underset{\sim}{x}$ for the step and $\underset{\sim}{y} \stackrel{\text{def}}{=} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^+) - \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x})$ for the change in the gradient. A symmetric matrix $\mathbf{B}$ satisfies the **secant condition** [2, Ch. 6] for $(\underset{\sim}{s}, \underset{\sim}{y})$ when

$$\mathbf{B}\underset{\sim}{s} = \underset{\sim}{y}.$$

Example 0.1 (The Hessian of a quadratic function satisfies the secant condition)**.** Let $f(\underset{\sim}{x}) = \frac{1}{2}\underset{\sim}{x}^\top \mathbf{Q}\underset{\sim}{x} - \underset{\sim}{b} \cdot \underset{\sim}{x}$ with $\mathbf{Q}$ symmetric. Then $\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) = \mathbf{Q}\underset{\sim}{x} - \underset{\sim}{b}$, so for any $\underset{\sim}{x}$ and $\underset{\sim}{x}^+$

$$\begin{aligned} \underset{\sim}{y} &= (\mathbf{Q}\underset{\sim}{x}^+ - \underset{\sim}{b}) - (\mathbf{Q}\underset{\sim}{x} - \underset{\sim}{b}) && \text{(gradient at each point)} \\ &= \mathbf{Q}(\underset{\sim}{x}^+ - \underset{\sim}{x}) && \text{(collect terms)} \\ &= \mathbf{Q}\underset{\sim}{s}. && \text{(definition of } \underset{\sim}{s}) \end{aligned}$$

So the Hessian $\mathbf{Q}$ of the quadratic function satisfies the secant condition for every step. For a general f, $\underset{\sim}{y} \approx \mathbf{H}_f \underset{\sim}{s}$ holds for a short step, with $\mathbf{H}_f$ the Hessian of f near the step, which is why a matrix that satisfies the secant condition is a reasonable stand-in for the Hessian.

Definition 0.2 (Quasi-Newton method)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, let $\underset{\sim}{x}^{(0)}$ be a starting point, and let $\mathbf{H}^{(0)}$ be a positive definite matrix. A **quasi-Newton method** [2, Ch. 6] produces, for $t = 0, 1, 2, \ldots$,

$$\underset{\sim}{x}^{(t+1)} \stackrel{\text{def}}{=} \underset{\sim}{x}^{(t)} - \alpha^{(t)} \mathbf{H}^{(t)} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}),$$

where $\alpha^{(t)} > 0$ is a step length chosen by a line search, and $\mathbf{H}^{(t+1)}$ is a symmetric positive definite matrix whose inverse satisfies the secant condition, that is, $\mathbf{H}^{(t+1)}\underset{\sim}{y}^{(t)} = \underset{\sim}{s}^{(t)}$, with $\underset{\sim}{s}^{(t)} = \underset{\sim}{x}^{(t+1)} - \underset{\sim}{x}^{(t)}$ and $\underset{\sim}{y}^{(t)} = \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t+1)}) - \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)})$.

Example 0.2 (One quasi-Newton step on a quadratic function)**.** Take $f(\underset{\sim}{x}) = \frac{1}{2}\underset{\sim}{x}^\top \mathbf{Q}\underset{\sim}{x} - \underset{\sim}{b} \cdot \underset{\sim}{x}$ with $\mathbf{Q} = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}$ and $\underset{\sim}{b} = (1, 2)^\top$ (Example 0.1), $\underset{\sim}{x}^{(0)} = \underset{\sim}{0}$ and $\mathbf{H}^{(0)} = \mathbf{I}_2$. For a quadratic function, the step length that minimizes f along a direction $\underset{\sim}{d}$ is $\alpha = -\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) \cdot \underset{\sim}{d} / \underset{\sim}{d}^\top \mathbf{Q}\underset{\sim}{d}$.

```
q_entries <- c(4, 1, 1, 3)
Q <- matrix(q_entries, nrow = 2)
b <- c(1, 2)
x0 <- c(0, 0)
H0 <- diag(2)
q_x0 <- drop(Q %*% x0)
grad0 <- q_x0 - b
d <- -drop(H0 %*% grad0)
q_d <- drop(Q %*% d)
step <- -sum(grad0 * d) / sum(d * q_d)
x1 <- x0 + step * d
q_x1 <- drop(Q %*% x1)
grad1 <- q_x1 - b
f0 <- 0.5 * sum(x0 * q_x0) - sum(b * x0)
f1 <- 0.5 * sum(x1 * q_x1) - sum(b * x1)
s <- x1 - x0
y <- grad1 - grad0
```

The step length is 0.25, which moves the iterate to $\underset{\sim}{x}^{(1)} = (0.25, 0.5)$ and lowers f from 0 to -0.625. The difference of the gradients, $\underset{\sim}{y}^{(0)} = (1.5, 1.75)$, is $\mathbf{Q}\underset{\sim}{s}^{(0)}$, which is the secant condition for a quadratic function.

ℹ Remark (Why a quasi-Newton method is used)

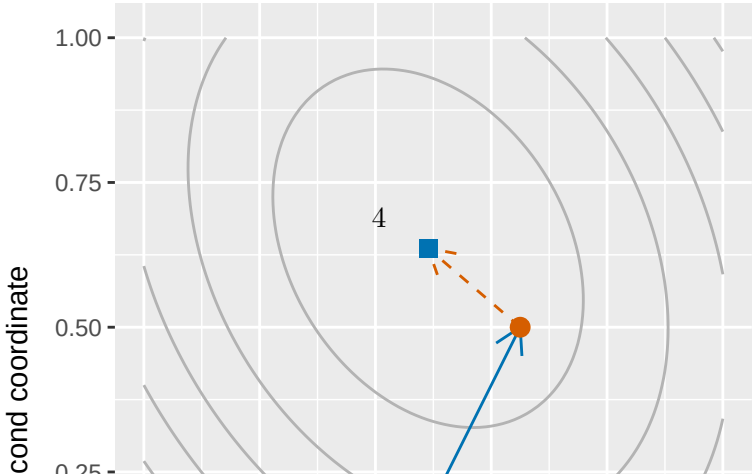
Remark 0.1 (Why a quasi-Newton method is used). Because $\mathbf{H}^{(t)}$ is positive definite, $-\mathbf{H}^{(t)}\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)})$ is a descent direction, so, when the gradient is not $\underset{\sim}{0}$, a line search finds a step that decreases f. Newton's method needs the Hessian and the solution of a system of equations at every iteration, which costs about p^3 operations for a dense $p \times p$ Hessian, and requires the formulas for its entries. A quasi-Newton method needs only the gradient and an update of $\mathbf{H}^{(t)}$ costing about p^2 operations. Near a minimizer the method speeds up superlinearly: faster than gradient descent, slower than Newton's method [2, Ch. 6]. Replacing $\mathbf{H}^{(t)}$ by $\mathbf{I}_p$ in the step formula gives gradient descent.

Definition 0.3 (BFGS update (Broyden-Fletcher-Goldfarb-Shanno))**.** Let $\mathbf{H}$, an approximation of the inverse Hessian, be a $p \times p$ positive definite matrix, and let $\underset{\sim}{s}, \underset{\sim}{y} \in \mathbb{R}^p$ satisfy $\underset{\sim}{y} \cdot \underset{\sim}{s} > 0$. With $\rho \overset{\text{def}}{=} 1/\underset{\sim}{y} \cdot \underset{\sim}{s}$, the **BFGS update** [2, Ch. 6] of $\mathbf{H}$ is

$$\mathbf{H}^{+} \overset{\text{def}}{=} (\mathbf{I}_p - \rho \underset{\sim}{s}\underset{\sim}{y}^{\top})\mathbf{H}(\mathbf{I}_p - \rho \underset{\sim}{y}\underset{\sim}{s}^{\top}) + \rho \underset{\sim}{s}\underset{\sim}{s}^{\top}.$$

R

```
qn_x_qn <- seq(-0.4, 0.6, length.out = 101)
qn_y_qn <- seq(-0.2, 1, length.out = 101)
qn_grid_qn <- tidyr::expand_grid(x = qn_x_qn, y = qn_y_qn)
qn_gx_qn <- qn_grid_qn$x
qn_gy_qn <- qn_grid_qn$y
qn_quad_qn <- 0.5 * (4 * qn_gx_qn^2 + 2 * qn_gx_qn * qn_gy_qn + 3 * qn_gy_qn^2)
qn_lin_qn <- qn_gx_qn + 2 * qn_gy_qn
qn_grid_qn$z <- qn_quad_qn - qn_lin_qn
qn_star_qn <- c(1, 7) / 11
qn_start_qn <- tibble::tibble(x = 0, y = 0)
qn_next_qn <- tibble::tibble(x = 0.25, y = 0.5)
qn_min_qn <- tibble::tibble(x = qn_star_qn[1], y = qn_star_qn[2])
qn_step_qn <- tibble::tibble(x = 0, y = 0, xend = 0.25, yend = 0.5)
qn_rest_qn <- tibble::tibble(
  x = 0.25, y = 0.5, xend = qn_star_qn[1], yend = qn_star_qn[2]
)
qn_len_qn <- grid::unit(0.15, "inches")
qn_arrow_qn <- grid::arrow(length = qn_len_qn)
qn_aes_qn <- ggplot2::aes(x, y)
qn_seg_qn <- ggplot2::aes(xend = xend, yend = yend)
ggplot2::ggplot(qn_grid_qn, qn_aes_qn) +
  ggplot2::geom_contour(ggplot2::aes(z = z), colour = "grey70", bins = 12) +
  ggplot2::geom_segment(
    data = qn_step_qn, mapping = qn_seg_qn, colour = "#0072B2",
    arrow = qn_arrow_qn
  ) +
  ggplot2::geom_segment(
    data = qn_rest_qn, mapping = qn_seg_qn, colour = "#D55E00",
    linetype = "dashed", arrow = qn_arrow_qn
  ) +
  ggplot2::geom_point(data = qn_start_qn, colour = "#0072B2", size = 3) +
  ggplot2::geom_point(data = qn_next_qn, colour = "#D55E00", size = 3) +
  ggplot2::geom_point(
    data = qn_min_qn, colour = "#0072B2", size = 3, shape = 15
  ) +
  ggplot2::coord_equal() +
  ggplot2::labs(x = "first coordinate", y = "second coordinate")
```



Example 0.3 (A BFGS update from the quasi-Newton step)**.** Continue Example 0.2 with $\mathbf{H} = \mathbf{I}_2$, $\underset{\sim}{s} = \underset{\sim}{s}^{(0)}$, and $\underset{\sim}{y} = \underset{\sim}{y}^{(0)}$.

```
rho <- 1 / sum(y * s)
left <- diag(2) - rho * s %*% t(y)
H1 <- left %*% H0 %*% t(left) + rho * s %*% t(s)
H1
#>         [,1]   [,2]
#> [1,]  0.6625 -0.425
#> [2,] -0.4250  0.650
drop(H1 %*% y)
#> [1] 0.25 0.50
```

Here $\underset{\sim}{y} \cdot \underset{\sim}{s} = 1.25 > 0$, so $\rho = 0.8$. The updated matrix satisfies $\mathbf{H}^{+}\underset{\sim}{y} = (0.25, 0.5)$, which is $\underset{\sim}{s} = (0.25, 0.5)$, and its smallest eigenvalue is 0.231204, so $\mathbf{H}^{+}$ is positive definite.

Remark (The curvature condition $\underset{\sim}{y} \cdot \underset{\sim}{s} > 0$)

Remark 0.2 (The curvature condition $\underset{\sim}{y} \cdot \underset{\sim}{s} > 0$). If the strict inequality of Theorem 4 in Convexity in Several Variables holds for distinct points, then $\underset{\sim}{y} \cdot \underset{\sim}{s} > 0$ for $\underset{\sim}{s} = \underset{\sim}{x}^{+} - \underset{\sim}{x} \neq \underset{\sim}{0}$ and $\underset{\sim}{y} = \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{+}) - \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x})$. Add the versions at $(\underset{\sim}{x}^{+}, \underset{\sim}{x})$ and at $(\underset{\sim}{x}, \underset{\sim}{x}^{+})$:

$$f(\underset{\sim}{x}^{+}) + f(\underset{\sim}{x}) > f(\underset{\sim}{x}) + f(\underset{\sim}{x}^{+}) + \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) \cdot \underset{\sim}{s} - \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{+}) \cdot \underset{\sim}{s},$$

so $0 > -\underset{\sim}{y} \cdot \underset{\sim}{s}$. For a general f, a line search that enforces a curvature condition on the step length (the Wolfe conditions [2, Ch. 3]) guarantees $\underset{\sim}{y} \cdot \underset{\sim}{s} > 0$. Such a step can be found along a descent direction when f cannot keep decreasing forever.

Definition 0.4 (Limited-memory BFGS (L-BFGS))**. Limited-memory BFGS** (**L-BFGS**) [2, Ch. 7] is the quasi-Newton method that never forms $\mathbf{H}^{(t)}$. It stores only the m most recent pairs $(\underset{\sim}{s}^{(j)}, \underset{\sim}{y}^{(j)})$, with m small, such as 5 or 20, and obtains $\mathbf{H}^{(t)} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)})$ implicitly, through a few vector operations, as if the BFGS update for those m pairs were applied to a simple starting matrix, such as a multiple of $\mathbf{I}_p$.

Example 0.4 (L-BFGS applies the update without forming the matrix)**.**
Continue Example 0.3 with $m = 1$ stored pair $(\underset{\sim}{s}^{(0)}, \underset{\sim}{y}^{(0)})$ and starting matrix $\mathbf{I}_2$. Below, $\mathbf{H}^{+} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(1)})$ is formed from vectors alone, starting from `H0`, then compared with the matrix product.

```
g1 <- grad1
alpha <- rho * sum(s * g1)
q <- g1 - alpha * y
r <- drop(H0 %*% q)
beta <- rho * sum(y * r)
r <- r + s * (alpha - beta)
h1_g1 <- H1 %*% g1
from_matrix <- drop(h1_g1)
tibble::tibble(from_vectors = r, from_matrix = from_matrix)
#> # A tibble: 2 x 2
#>   from_vectors from_matrix
#>          <dbl>       <dbl>
#> 1        0.438       0.438
#> 2       -0.375      -0.375
```

The two columns agree to within 5.6e-17, although the first never forms the matrix. BFGS stores a dense $p \times p$ matrix and spends about p^2 operations per iteration; L-BFGS stores $2m$ vectors with p entries and spends about mp operations per iteration, so it remains usable when p is in the millions.

[1] H. Zhou, "Quasi-Newton methods." Lecture notes for Biostat 257, Statistical Computing, University of California, Los Angeles, 2025. Accessed: Oct. 26, 2026. [Online]. Available: https://ucla-biostat-257.github.io/2025spring/slides/24-quasinewton/quasinewton.html

[2] J. Nocedal and S. J. Wright, *Numerical optimization*, 2nd ed. in Springer series in operations research and financial engineering. Springer, 2006. doi: 10.1007/978-0-387-40065-5.